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SDS210

Does median income affect the distribution and strength of urban heat island effects in the city of Zurich?

Introduction:
Residents of cities often experience higher temperatures than those in the suburbs or rural areas. This is known as the urban heat island effect and occurs mainly because dense urban areas store heat more effectively and have decreased ventilation and transpiration compared to vegetation (Mohajerani et al., 2017). The cooling effect of vegetation is well known, however it is often unequally spread across socioeconomic boundaries such as income (Dialesandro et al., 2021). In this study, I use time series data from the Landsat satellite to identify how vegetation cover and surface temperatures have changed in Zurich over time. I then map it across districts with varying income levels to see if urban heat inequality is also present in Switzerland and if it has changed over time.

Methods:
I used four spatial datasets for this study. The triannual Landsat satellite rasters from 1985-2024, the vector polygon layer of the city of Zurich and its districts, the median annual income of each district from 1999-2023, and lastly a vector polygon of the zoning types in Zurich. I calculated the normalized difference vegetation index (NDVI) from the red and near infrared bands of the Landsat data. Then, I merged the district and income layers to spatially map the median income of each district per year. Afterwards, I used the zoning data to mask all water bodies and forests as these would have a significant effect on NDVI and land surface temperature (LST) measurements. I used the xarray spatial zonal stats tool to combine the district polygon layer with the satellite raster and calculate mean NDVI and LST values for each district over time, and I mapped the change over time to see which districts changed the most. Finally, I plotted all mean district LST and NDVI values against each other to identify how vegetation cover affects surface temperatures and tested for correlation in change of NDVI or LST with change in income.

Results:
There is no correlation between either surface temperature and income ($r=-0.076$, $p=0.461$) or vegetation cover and income ($r=0.061$, $p=0.555$) across Zurich districts. Over the entire dataset the inner city districts near the main station consistently had the highest urban heat island effects, whereas those further out were considerably lower. Change in income was positively correlated with change in NDVI ($r=0.278$, $p=0.011$) and negatively with LST ($r=-0.217$, $p=0.048$). This trend can be best seen in district 5 which had the highest income increase and second highest NDVI increase (Figure 1).

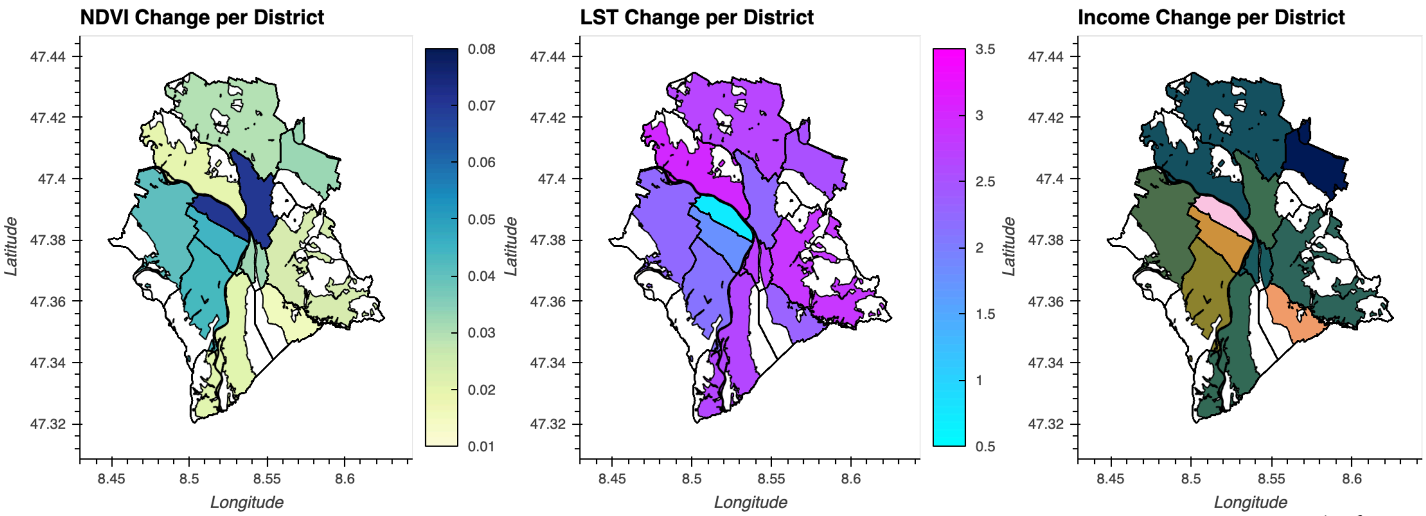

Figure 1: Total change in NDVI, LST and income per district.

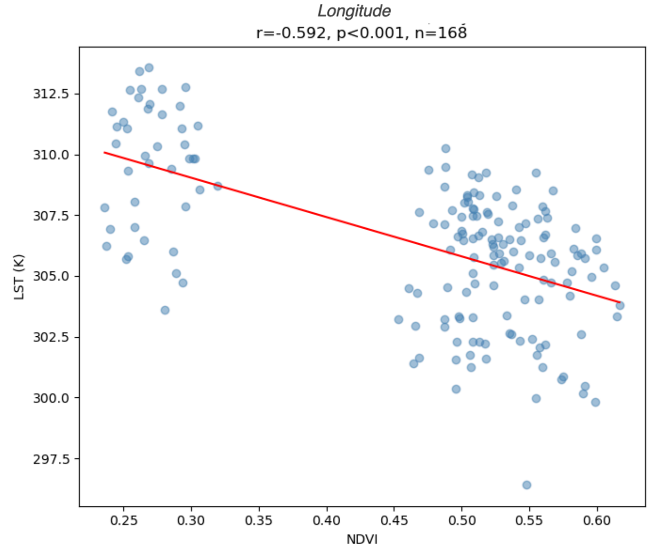

Figure 2: NDVI / LST correlation plot across all districts and years. The negative correlation between LST and NDVI values is clearly shown.

Discussion:

The urban heat island effects were strongest in the inner city districts across all years, especially those by the main train station. As income is not correlated with this heat island effect there it does not represent a factor in the heat inequality occurring in Zurich. The heat island distribution can likely best be explained by population density and the historical development of the city. However, as shown in figure 1, districts which increased the most in income also had higher changes in NDVI, so while heat inequality is not generally a concern, there may be concern for stagnant districts to suffer more in the future. Overall though, the changes in NDVI and LST across districts are small, and the districts with lowest LST remained the same, showing that environmental variables across Zurich are stable. Finally the cooling effect of vegetation is clear as NDVI negatively correlated with LST (Figure 2.)

The largest analytical challenge for me was figuring out how to create meaningful NDVI and LST district averages. Many of the district boundaries include sections of lake or river, these are not areas citizens spend most of their time in and are irrelevant for our study and they heavily skew the NDVI and LST values. The same logic counts for the forests in the city, I wanted to analyze the areas that citizens actually live and work in. To define all forest and water areas was difficult, at first I thought I could mask any values above or below a certain threshold but quickly realized that this may remove vegetation that is not solely forest or areas that are water. Instead, I used the map of the zoning regulations which included forest and water zones and created a mask for these. This allowed me to calculate meaningful district averages. This was also the most technically challenging step as combining the raster bands from rioxarray with vector polygons from geodataframes was not straightforward. Thankfully, calculating means per district boundaries for zonal statistics is a common step in spatial data analysis and there was already a package for such a task. Using “geocube” to rasterize my polygon data and “xrspatial” to compute zonal statistics I was able to create maps of district averages of NDVI and LST. The main limitations of this project are my own weaknesses with statistics, the analyses are rather simple and reduce the millions of data pixels to 12 values per year. The aggregations could have been done on smaller levels such as city neighborhoods or even done analysis on all pixel changes to highlight specific areas of high change. These are ideas I would like to explore more in the future as my analysis only captures broader scale trends.

AI Statement: *I used a large language model as an assistance tool during the preparation of this project to help debug code for data processing, help with the implementation of the "xrspatial" package and aid in the creation of my correlations. Apart from the uses listed above, no AI tools were used in the preparation of this submission, and all final logic and writing was verified by me.*

Repository Link: https://github.com/staehelinhenrimax-droid/Zurich_Heat/tree/main

References

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Mohajerani, A., Bakaric, J., & Jeffrey-Bailey, T. (2017). The urban heat island effect, its causes, and mitigation, with reference to the thermal properties of asphalt concrete. *Journal of Environmental Management*, *197*, 522–538. https://doi.org/10.1016/j.jenvman.2017.03.095